ELECT MAINT Department
SIMHADRI

Note No :1

Ref:SMTPP/ELECT MAINT/2024-25/LMIEMHT1/607987 **Date:**22/10/2024(10:44:20)

Subject:LMI for Guidelines on HT Switchgear Maintenance Practices

NTPC CC-OS issued Operation guidance note for "Guidelines on HT Switchgear Maintenance Practices" vide

OGN No: OGN-OPS-ELEC-044.

New LMI was created vide: LMI/OGN/OPS/ELEC/044 to adopt the OGN issued above by CC-OS.

The LMI covers various guidelines for safe and reliable operation & maintenance of HT Switchgear.

LMI copy is atatched in pradip note file.The same may please be approved.

Put up for consideration and aproval of competent authority please.

Submitted by:

जॉनसन प्रसन्ना कुमार कालाडासी - वरिष्ठ प्रबंधक

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Note No:2

checked and forwarded for further approval

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Note No:3

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Note No:4

Please review.

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Note No:5

The LMI was reviewed and found in order.

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Note No:6

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Note No:7

LMI reviewed as per discussion.

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Note No:8

Forwarded for further approval

आगे की स्वीकृति के लिए भेजा गया

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Note No:9

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

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Note No:10

Submitted by:

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Note No:11

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On:05/11/2024(11:29:40) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIEMHT1/607987

Note No:12

LMI OGN OPS ELEC 044 Guidelines on HT switchgear Maintenance Practices_051124.pdf attached for review.

Submitted by:

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Note No:13

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Note No:14

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On:06/11/2024(18:14:52) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIEMHT1/607987

Note No:15

LMI modified as per standard format.put up for approval please.

Submitted by:

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Note No:16

Submitted by:

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Note No:17

It is new LMI Rev.0 is to be mentioned on every page along with LMI No. as per standard format

Submitted by:

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Note No:18

Necessary changes as mentioned above done. May please review and froward.

Submitted by:

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Note No:19

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Note No:20

submitted for review and approval please.

Submitted by:

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Note No:21

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Note No:22

LMI is in line with the OGN.

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Note No:23

Unauthorized Use Prohibited

Reviewed and submitted for approval

Submitted by:

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Note No:24

May Please be approved.

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Note No:25

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NTPC SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

GUIDELINES ON HT SWITCHGEAR MAINTENANCE PRACTICES

LMI/OGN/OPS/ELEC/044, Rev: 0

OCT - 2024

APPROVED FOR IMPLEMENTATION BY HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI Name	Guidelines on HT Switchgear Maintenance Practices
2	LMI No	LMI/ OGN/OPS/ELEC/044,REV:0
3	Reference Document	COS-OGN/OPS/ELEC/044,REV:0
4	Superseded Document	NIL
5	Prepared by	K.J.Prasanna Kumar Sr.Manager (EM)
6	Checked by	Satish Kumar Singh DGM (EM)
7	Checked by	A Bala Balaji DGM (EM)
8	Checked by	Tenneti Vijay Krishna HOD(EM)
9	Reviewed by	Om Prakash AGM(Maint)
10	Reviewed by	Prasanta Kumar Jena GM(O&M)
11	Approved by	Sameer Sharma HOP (Simhadri)

INDEX

1.0 INTRODUCTION

The switchgears are the source of auxiliary power system in a power plant. They are the reliable source of power to all units and station auxiliaries. The basic function of switchgear is to control the supply of electric power and to protect the equipment in the event of abnormal conditions. Hence, the switchgears must be reliable, safe, and adequate for continuous running of Generating Units.

Over the years, from time-to-time CC-OS has issued several guidelines for safe and reliable operation and maintenance of HT Switchgear. The guidelines have been issued in the form of IOMs, and technical compliance documents like OIN, OGN, OD etc. For quick and easy reference of the guidelines by station maintenance personnel, it has been decided to integrate the recommendations into a single document in the form of this OGN.

2.0 Superseded Document –Not Applicable

3.0 Records of revision – Not Applicable

4.0 Scope: Guidelines on HT Switchgear Maintenance Practices (3.3kV to 33kV)

5.0 GUIDELINES ON MAINTENANCE OF HT SWITCHGEAR

1. Only authorized personnel should be allowed to work on the MV (3.3/6.6/11/33 kV) switchgear.
2. Detailed Isolation and normalization procedure for each switchgear is to be prepared and the same are to be uploaded for the respective switchgear in the SAP system. The isolation procedure should mention the isolation from the remote end also. (If transformers are fed from the switchgear, then isolation from the LT side of the Transformer is to be ensured). The isolation of PT should also be done to avoid back feeding from PT secondary side. Isolation/normalization SOPs to be displayed at Switchgear rooms.
3. Prior to the start of maintenance work on any switchgear, a Job safety Analysis (JSA) is to be conducted based on actual site condition and identification of the hazards.
4. The isolation/normalization procedure and the job safety analysis are to be discussed in advance with the operation personnel (responsible for doing the isolation and normalization).
5. Arc flash suit and PPE must be worn during isolation and normalization of the MV switchgear.
6. After isolation of the switchgear and before start of the work, cross check the isolation and ensure zero voltage at PT panel, No LED should glow in all 3 phases at PT Panel, then a non-contact type of voltage detector is to be used for ensuring that the switchgear (where maintenance work is planned to be executed) is isolated from the live parts. If during isolation of any switchgear, the live part at some of the panel (like incomer/Tie) cannot be isolated, then proper locking

arrangement of the live panel (from both front/rear side) is to be done with provision of danger tags, so that inadvertent opening of the panel is avoided.

7. Proper identification markings on front and back side of panels to be done. Marking to include name of equipment, panel no. and cross marking across entire switchgear.
8. Cross marking will help to identify the correct panel cover.
9. Sealing of all switchgears to be ensured. It is to be ensured that there is no water ingress inside the bus duct and switchgear room. Daily walk down checklist must be followed, and corrections must be made by maintenance.
10. To prevent entry of rodents in switchgear, periodic pest control and hole sealing is to be carried out. Periodic checking of hole sealing to be done.
11. Proper illumination of the complete switchgear room is to be ensured.
12. Availability & working effectiveness of AHU/AWU must be ensured in Switchgear rooms.
13. There should be no water in the trenches below and humidity control is to be done through exchange of air.
14. Incomer/Bus Coupler/ Tie breaker live side marking on panel.
15. Switchgear SLD is to be displayed in the Switchgear Room.
16. MV Switchgear operation, maintenance, and safety training to be imparted to all the concerned persons.
17. Correct rating testing, grounding equipment and procedure to be always used.
18. An electrical safety audit may be carried out for all switchgears once in 3 years.
19. Periodic breaker OH is to be done after reaching requisite no. of operations. OEM's recommendations regarding the number of operations of breakers for maintenance are to be followed.
20. Periodic testing protective relays along with SCADA/DDCMIS Alarms/Events of relays, CT/PT Testing, checking of auxiliary relays, interlock, scheme check, wiring/cabling healthiness (IR & continuity), Auto changeover of Unit bus (11/6.6 kV) to station bus and vice versa, Changeover schemes checking of 3.3 kV switchgear as per LMI: COS- ISO-00/OD/OPS/ELEC/003.
21. Numerical Relay replacement guideline and maintenance is to be followed as per LMI: COS-ISO-00-OGN/OPS/ELEC/040
22. All the Electromechanical and Static relays of HT Switchgears are to be replaced with Numerical Relays.
23. Core Balance Current Transformers (CBCT) are being used in HT Switchgear should be circular type. There should be a single CBCT in each feeder. The window size of CBCT should be based on overall diameters of cables.
24. Physical segregation of power & control cable to be ensured. Armored cable one end earthing to be checked. Cable termination tightness and checking of gland is to be done.

6.0 HT SWITCHGEAR PROTECTION SCHEME

1. The numerical relay setting in motor feeders shall have 2 group sets. The first group set shall be active during running and second group set should be always active/ during starting of motor depending on numerical relay used.
2. In motor feeders
3. 86 Trip shall include – Differential, Locked Rotor, Time Delayed Over Current, Instantaneous Over Current, Earth Fault, Negative Sequence, Load Jam.
4. Non 86 Trip shall include – Thermal operation/Overload, Phase Under Voltage
5. The under voltage used in motor feeders should be 3-phase and interlocked with PT fuse fail in numerical relays. Internal blocking is to be enabled in control setting with Fuse Fail.
6. During slow bus changeover, Motor feeders should trip on under voltage after a time delay of 2 sec. then station tie incomer/ bus coupler will close after a delay of 2 sec on reaching no voltage in the bus. This will ensure complete disconnection of all motor feeders from the bus before reenergizing through standby source.
7. In transformer feeders

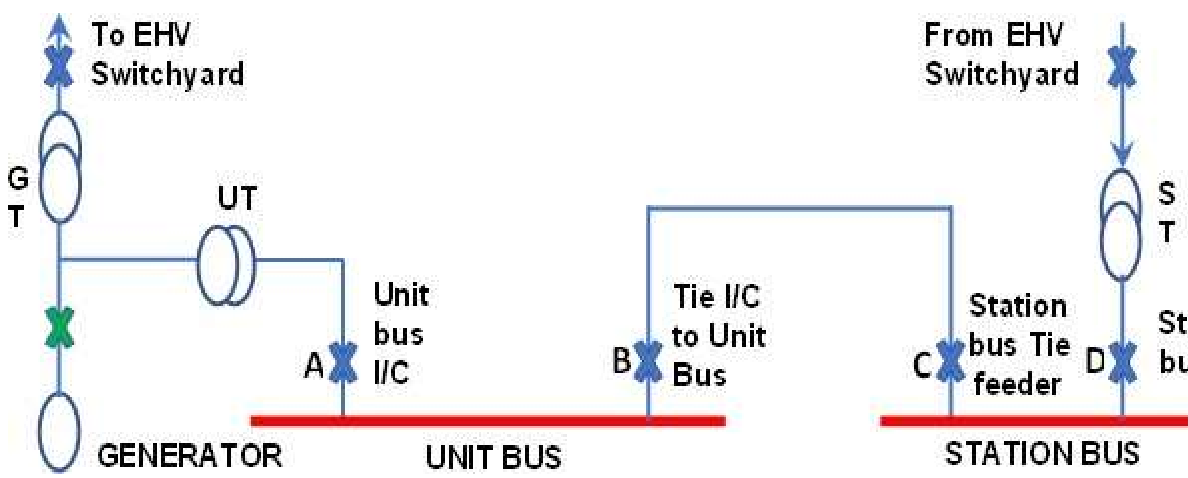
a. 86 Trip shall include – REF from downstream, Differential, Time Delayed Over Current, Instantaneous Over Current, Earth Fault, Buchholz trip, PRV trip.

b. Non 86 Trip shall include – Winding Temp high, Oil Temp high, Phase Under Voltage.

8. A numerical relay/master trip contact shall be used in breaker closing circuit which should be latched type from 86 Trip.
9. The electrical trip operated should be used in close command interlock of DDCMIS (C&I) logic to avoid inadvertent command by operation. All 86 (latched) and non 86 electrical protections shall be used for interlocking.
10. One EPB contact shall be provided in closing interlock. Other EPB contact shall be used directly in breaker tripping circuit.
11. Whenever the drive control is through DCS, one EPB contact shall be used for actuation of one interface relay or binary input of numerical relay which is to be provided in the respective switchgear relay panel. “EPB Operated” signal to DCS system is to be taken from the contact of the interface relay or binary output of numerical relay instead of the existing practice of directly from the EPB.
12. In reverse blocking scheme only instantaneous over current, instantaneous earth fault protection shall be provided from feeders to incomers/tie breaker/Bus coupler. The protection start should be used in numerical relay-based schemes.
13. Relay DC circuit with fuse may be shifted out from breaker DC circuit so that relay shall remain in service if the breaker DC is made off.

14. SOP shall be jointly prepared by EMD and Operation for the cases of switchgear breaker failure to trip.
15. The incomers of UT & ST should not trip on transient under voltage. Settings and delay to be checked accordingly.
16. Electrical tripping of any HT equipment should be thoroughly analyzed before giving clearance to operation. Disturbance records and relay event logs should be extracted immediately after the incident and are to be analyzed.
17. The breaker status is to be taken from a latched- type of Relay/ Contact with breaker in service position. (i.e., either in case of failure of DC supply or operation of the breaker in test position, the status of the breaker shall not change and cause an unwarranted tripping/other disturbance). If not, then the same is to be modified at the earliest opportunity.
18. In incomer of Unit Board: To avoid maloperation of 50N (Residual earth Fault), it is recommended to use the 51N pickup contact of Unit Transformer as an interlock for operation of 50N. Similarly, in tie of Unit Board & Station Board and Incomer of Station Board: To avoid maloperation of 50N (Residual earth Fault), it is recommended to use the 51N pickup contact of Station Transformer as an interlock for operation of 50N.
19. A debounce time of 20 milli seconds is to be provided to all binary inputs used for tripping or trip interlock application in Numerical relays to avoid spurious pick of binary inputs.
20. The reference calculation for high impedance restricted earth fault protection is attached as Annexure-1.

7.0 BUS CHANGE OVER SCHEME OF 11/6.6 KV UNIT SWITCHGEAR



<table>
<tr><th>SI No</th><th>Description</th><th colspan="2">Condition</th></tr>
<tr><td>1</td><td>Condition, with Unit in Running Condition</td><td colspan="2">i. Breaker “D” in CLOSED Condition
ii. Breaker “C” in CLOSED Condition
iii. Breaker “B” in OPEN Condition and remain Ready for AUTO CHANGEOVER
iv. Breaker” A” in CLOSED Condition</td></tr>
<tr><td rowspan="3">2</td><td rowspan="3">Change Over Logic</td><td>Condition-1
(Manual Change Over)</td><td>Manually operator can give the close command from DCS to B and automatically DCS releases trip to breaker A as soon as close feedback of B is received and vice versa</td></tr>
<tr><td>Condition-2
(Auto Fast Change Over)</td><td>With either Outage of Unit or in case of oil and Winding temperature Hi-Hi, Trip command is extended to Breaker: A” and Close Command is extended to Breaker “B”</td></tr>
<tr><td>Condition-2
(Auto Slow Change Over)</td><td>When the breaker A is open, and no volt of unit bus is operated with a delay of 2 sec then DCS release auto close command to breaker B</td></tr>
<tr><td>3</td><td>NOTE</td><td colspan="2">For all the above conditions, numerical relay at B checks for the following conditions.
i. Availability of synchronization condition
ii. Healthiness of the Breaker
iii. No Electrical Protection Operated</td></tr>
</table>

8.0 RECOMMENDATIONS FOR HT SWITCHGEAR

1. Visual inspection of the protective devices for any abnormal condition and Visual Inspection of LEDs and resetting after due diligence on daily basis.
2. All differential protection spill currents shall be checked at least once in a fortnight, and it should be less than 10% of Id min setting. Any higher value should be investigated.
3. The healthiness of HMI Station being used for SCADA should be checked on a daily basis. Ensuring all relays should communicate to HMI Station.
4. Monitoring of 220V DC Voltage in All DCDB (Positive to Earth & Negative to Earth) on daily basis for ensuring that there is no DC Earth Fault. If any is to be identified and rectified at the earliest.
5. Ensuring DC Earth Fault to Alarm/SOE in DCS being checked on quarterly basis. Ensurethe DC voltages (P-E & N-E) are available in trends at DCS.
6. Inspection of the Breathers provided in the Bus duct (replacement of silica gel if required) on monthly basis.
7. Thermal Scanning of switchgears on monthly basis.
8. Monitoring of Healthiness of the Space Heaters by monitoring the Space Heater Currentof the Breaker Panels on monthly basis.
9. While working on the Live panel/system (secondary circuit), SOP & JSA are to be prepared. The matter is to be discussed with senior people before executing the work.

9.0 TESTING OF SWITCHGEAR COMPONENTS

a. CURRENT TRANSFORMER (CT) OF INCOMER & FEEDERS

i. IR Value measurement of primary to secondary terminal (Bar mounted CT), primary to earth, Secondary to earth and secondary to secondary terminal.
ii. IR test of Critical feeders i.e. Incomer/Tie/Bus Coupler/ID/PA/FD/MDBFP/UAT Feeders must be done during Overhauling.
iii. Ratio Test
iv. Ratio test of Critical feeders i.e. Incomer/Tie/Bus Coupler/ID/PA/FD/MDBFP/UAT Feeders must be done during Overhauling.
v. Measurement of CT Secondary Resistance
vi. Checking of Knee Point Voltage of CT (applicable for PS CLASS CT)
vii. Polarity Test

b. POTENTIAL TRANSFORMER (LINE AND BUS PT)

i. IR value of Primary to Earth, Secondary to Earth, Primary to Secondary
ii. Ratio test of PT (Rated voltage or 5 kV whichever is lower)

iii. Magnetizing current of PT (in steps up to Rated voltage or 5 kV, whichever is lower)
iv. Checking of the Resistance of PT Primary Fuse and replacement of the PT primary Fuse every 10 years.
v. Checking of PT secondary circuit fuses
vi. SURGE ARRESTER IN MOTOR FEEDERS

a. Cleaning and inspection of Surge Arrester

b. IR value of Surge Arrester

c. PROTECTION RELAY & CONTROL CUBICLE

a. Disconnection of CT and PT Secondary terminals from Relay and Meter in CT and PT Terminal Blocks of respective Cubicle.
b. Checking of the operation of protective Relays by secondary injection from CT terminal blocks of the respective cubicle.
c. Checking of reverse blocking signals during testing of Over current, Earth fault and Differential protection of each feeder to Incomer/Tie/Bus Couplers of corresponding switchgear section
d. Checking Relay Binary Inputs, Binary Outputs and LED's during testing of Numerical relays.
e. Checking of the tripping of the Breaker by operation of the protection Relay.
f. Checking of the Alarm and SOE points up to the TB from where C&I interface is connected (from TB to DCS system shall be checked jointly with operation)
g. Checking of the display of electrical parameters (current, voltage, kW etc.) in the meters/Relays/Transducers by injecting voltage and current from the terminal block (starting point of the CT & PT of the respective control cabinet)
h. Ensure all the settings of Relays are restored to recommended value (in case any changes done during testing)
i. Numerical relay DR triggering on Breaker Close, Trip condition, C&I Open/ Close command, 86 & Non 86 Trips to be ensured.
j. Checking of the CT secondary side and Relay side Resistance (by multi-meter)
k. Connection of the link of the CT & PT. After connection of disconnecting type CT TB link, resistance measurement across connected link is be done to ensure link is properly intact.
l. Ensuring the healthiness of indicating Lamps of the panel.
m. Check the healthiness of diode bridge circuit provided.
n. Checking of changeover logics/circuits

d. **CABLES**

i. HT CABLES

1. IR value of Cables.
2. Capacitance & Tan δ and tip up value to be measured for all the HT cables of motor feeders and Critical Transformer feeders during overhauling. The base line data is to be generated for the first time and the result should be compared with the earlier test results.

ii. CRITICAL CONTROL/PROTECTION/CT SECONDARY CABLES

1. The IR value of all Current transformer's secondary cables related to unit protection & critical applications are to be measured and single point earthing of neutral to be confirmed. Loop resistance of secondary circuit to be measured.
2. The IR value of Critical control cables relating to tripping of unit / stations to be measured and the values are to be recorded. Physical inspection to be carried out to assess the condition of the cables / joints. Joint-less / single run cables to be laid for critical applications.

e. BREAKER TESTING

1. IR Value measurement of Breaker Poles (Pole to Earth & Pole to Pole of Incoming & Outgoing side)
2. Healthiness checking of SF6 bottle (by checking the limit switch provided for the same)
3. Vacuum Healthiness test of Breaker by application of rated voltage(applicable for VCB only)
4. Breaker Timing (Close, Open & Close-Open)
5. Contact resistance Measurement of Breaker Main Contacts

10.0 CALCULATION OF SETTING FOR REF PROTECTION OFTRANSFORMER

Inputs required:

a) Voltage rating at nominal tap
b) Current at nominal tap
c) MVA rating of winding (Value X)
d) Current Transformer (CT) ratio (CTR) used for phase and neutral and CT resistances(Rct).
e) Knee point of individual CT's (Vk)
f) % Impedance (Value –Z) at (Y) MVA base which is mentioned on transformer nameplate.
g) Cable length between CT and point where all REF CTs are paralleled (L).

Calculations:

a) Operating current/ sensitivity (Is) – 10% of rated current in CT secondary terms(generally kept 0.05 or 0.1)
b) Maximum through fault current (Ikmax) – [100*rated current / (Z*X/Y)]
c) Value of Rct+2*RL = Rct + (2*8.71* L), this shall be calculated for individual CTs.
d) U = [Ikmax / CTR] * (Rct + 2*RL). The value Rct+2RL should be maximum out of values calculated for all CT's used in REF.
e) Stability voltage desired for through fault stability (Us)= U<Us<(Vk/2)
f) A factor of safety (f) is considered between 1.0-2.0 while deciding Us
g) Us = f*U. It is recommended to keep f=1.5 if (f*U) <(Vk/2) else it can be kept between 1.2-1.4.
h) Stabilizing resistance = Us/Is
i) A nonlinear resistor (Metrosil) is required to limit the CT output voltage under an internal fault if the magnitude of peak voltage (VP) is higher than 2 kV.

Vp=2*Sqrt (2Vk*(Vf - Vk))

VF = (IFint/CTR) * (RCT + 2RL + RS) IFint shall be 50/63kA for grid side REF and Ikmax for LV side REF of ST/UT transformers where on LV there is no grid source.

Note:

1) It is to be ensured that in REF all CTs should be PS class and shall have same magnetizing characteristic which should be tested during every overhaul of Transformer.

2) Other transformer protection relays should be configured such that it will trigger DR during REF relay operation (may be done through any binary input), so that phase currents are available during fault.

3) DR pre trigger and post trigger duration shall be at least 0.5 sec.

4) Metrosil (Non-linear resistance) should be properly connected and checked for leakage current at (Vk/2).

5) After every tripping on transformer stabilizing resistance should be checked for healthiness.

11.1 Protocol for changeover of switchgear (11unit switchgear)

A station specific procedure is to be prepared for checking the changeover of switchgear in detail:

1. UT breaker Close to Tie Breaker Trip from Remote (DCS/ECD)
2. Tie Breaker Close to UT Breaker Trip from Remote (DCS/ECD)
3. Fast Auto Changeover checking

   a.Auto closing of Tie Breaker & Auto Tripping of UT Incomer Breaker onoperation of Winding and Oil temperature High-High of respective UT/UAT

   b.Auto closing of Tie Breaker & Auto Tripping of UT Incomer Breaker on operation of Class-A Protection of GRP
4. Slow Auto Change over checking

   a.Auto closing of Tie Breaker through slow changeover (No Volt with a time delay of 02 seconds), in case of failure of the fast changeover. To check Slow Auto Change Over scheme, the fast change over scheme is to be kept disabled.

MPORTANT NOTE- DURING CHANGE OVER CHECKING, THE SOE IS TO BE CHECKED FOR CORRECTNESS OF TIMING OF CLOSING AND TRIPPING OF BREAKER. COPY OF SOE IS TO BE KEPT ALONG WITH PROTOCOL.

11.2 Protocol for SOE& alarm for individual breaker control

a. ON/OFF Status of Incomer/Tie breakers in SOE (1 milli sec time scan) in DCS

b. ON/OFF feedback Status of Feeder (Motor/Transformer) in Alarm/SOE

c. Breaker Disturbance (in case of Control supply/associated system failure)

d. Breaker Not in Remote (breaker is not in service, or the selection of switchgear isnot in remote)

e. Overload Alarm of Motor Feeders in DCS system

f. Electrical protection operated (any electrical protection operated) in DCS.

g. Trip Circuit Supervision Alarm in DCS System
h. Relay unhealthy Alarm in DCS System
i. EPB operated Alarm for Motor Feeders in DCS

11.3 Protocol for SOE& alarm for common switchgear section checks

Bus under Voltage Alarm (70% 2 Sec) in DCS SOE/Alarm. It should be 3 phase undervoltage.

a. Bus No Volt Alarm (less than 20 %) in DCS SOE/Alarm. It should be 3 phases No -Volt.
b. UT-ST BUS-A/B changeover trouble status in DCS SOE/Alarm
c. Reverse blocking active Alarm/SOE in DCS (Reliability of Switchgear protection system to avoid incomer breaker trip during a feeder fault)
d. Any Breaker Fault in the respective section (Due to the limited no. of operator console in the control room, all the electrical breakers status cannot be displayed and difficult to be monitored. In case of any disturbance/outage of any feeder breaker (through DCS control), the same may go unnoticed and create an emergency. To take care of this, the signal of breaker fault/switchgear disturbance of all the breakers of that section can be used in OR to get an alarm for the operator, so that he can check further for identification of the faulty feeder and remedial measures can be taken)
e. Respective section DC supply – 1/2 Fail (The DC supply any source fail alarm is to be taken after the switchgear panel Fuse)
f. PT Fuse fail status in DCS Alarm

NOTE - There are three following types of schemes adopted in NTPC.

- ✓ One is through diode coupling – where both sources are always extended to individual feeders and in case of interruption of one DC source, the second source will automatically take over and there will be practically no interruption of Control Supply to any of the feeders/Incomers/Tie breaker of that section. It has its own disadvantage (in case of failure of Diode block, both the source shall fail & it is time taking to identify the faulty diode block)
- ✓ Changeover through Contactor – Two sources of DC supply are available in a particular section. Contactor logic is incorporated. In this case with interruption of one DC supply source, the other source shall come into service and in this case, there will be interruption of Control supply for around 200 milli seconds due to inherent delay of the pickup of the contactor.
- ✓ Manual intervention – Two sources are available in a particular section. There is a manual selection of DC supply source and in case of failure of one DC supply source, the operator must change the selection from the earlier selected source to the available source.

The above three schemes as per station applicability are to be jointly checked with operation.

11.4 Protocol for emergency manual tripping of breakers hecks

Emergency Manual Tripping of MV Breakers (11/6.6/3.3kV) are to be jointly checked with operation and joint protocol with operation is to be made.

12.0 MAINTENANCE SCHEDULES OF HT SWITCHGEAR

Sl. No.	EQUIPMENT	WORK TO BE CARRIED OUT	FREQUENCY
HT SWITCHGEAR			
01	SWITCH GEAR PANEL	Inspection, cleaning, sealing & maintenance of all HT panels, tightness checking of all control circuits and earthing connections.	During OH or Once in Every 02 year (if not related to Unit OH)
		Breaker maintenance & testing.	During OH or Once in Every 02 year (if not related to Unit OH)
		Secondary injection testing of Protective Relaysalong with SCADA/DDCMIS Alarms/Events of relays in Switchgear.	During OH or Once in Every 02 year (if not related to Unit OH)
		Checking of auxiliary relays, interlock, schemecheck, wiring/cabling healthiness (IR & continuity) of switchgear protections.	During OH or Once in Every 02 year (if not related to Unit OH)
		Auto changeover of Unit bus (11/6.6 kV) tostation bus and vice versa	During OH
		Changeover schemes checking of 3.3 kVswitchgears	During OH or Once in Every 02 year (if not related to Unit OH)
		Panel Meters- Inspection, Cleaning & Testing bysecondary injection.	During OH or Once in Every 02 year (if not related to Unit OH)
		Inspection, cleaning & maintenance of bus bars,support insulators, IR measurement of bus, milli- volt drop test of bus.	During OH or Once in Every 02 year (if not related to Unit OH)

		CT and PT Testing.	During OH or Once in Every 02 year (if not related to Unit OH)
		Surge Arrestor testing in Motor feeder	During OH or Once in Every 02 year (if not related to Unit OH)
		Complete Breaker Overhaul	Every Four Years
		Capacitance & Tan δ and tip up value to be measured for all the HT cables of motor feeders and Critical Transformer feeders	During OH or Once in Every 02 year (if not related to Unit OH)

13.0 PROBABLE CAUSES OF UNIT OUTAGE DUE TO HT SYSTEM AND ACTIONPLAN

SI No	Reason of Outages	Action Plan for Reliability
01	SPBD Related Outage i.e. Moisture ingress, Seal off bushing damage, poor sealing of SPBD enclosure etc.	▪ **Monthly** ➢ Thermal Scanning of the Bus duct ➢ Inspection of the Breathers provided in the Bus duct (replacement of silica gel if required) ➢ Monitoring of Healthiness of the Space Heaters by monitoring the Space Heater Current. ▪ **Overhauling** ➢ Inspection of the Bus duct and the Seal off Bushing including Flexible connection ➢ Sealing of the Bus duct Joints by additional Sealant/Aluminum Tape ➢ Air tightness test of the Bus duct by making special arrangements. ➢ Making the Space Heaters available ➢ Application of Voltage for ascertaining the Healthiness of the Bus duct (110 % of the working Voltage)

02	Flash Over in the Incomer/Tie/Bus CouplerBreaker	▪ **Monthly** ➢ Thermal Scanning of the Incomer Breaker Panel. ➢ Monitoring of Healthiness of the Space Heaters by monitoring the Space Heater Current. ▪ **Overhauling** ➢ Inspection and Overhauling of the Breaker ➢ Measurement of the contact swipe ▪ **Temperature Monitoring System is to be provided for Breaker contacts.**
03	Failure of Change Over logic	▪ **Overhauling** ▪ Auto change over logic is to be jointly checked with operation. ▪ Complete functional checks of auto changeovers are to be checked. ▪ The SOE is to be checked for evaluation of the timing of thebreaker operation. ▪ Protocol is prepared for the auto change over
04	Failure of Protective Components (CT, PT, LA)	▪ **Fortnightly** ➢ Wherever Diff Relays are provided, the spill currents are to be measured. ▪ **Overhauling** ➢ Testing of the CT (IR Value, CT secondary Resistance, Ratio, Knee Point) ➢ Testing of the PT (IR Value, ratio & Mag Current) including measurement of resistance of PT primary Fuses and secondary fuses. ➢ Tightness checking of the CT/PT secondary Cables (All CT/PT secondary connections should be of ring terminal type) and should be flexible copper of 2.5 sq.cm ➢ Proper cable routing of the CT/PT secondary cables to prevent mechanical damage. ➢ IR value and High Voltage checking of Surge Arresters (at 110 % of the working Voltage)

05	Flash Over or Cable Failure	▪ **Monthly** ➢ Thermal Scanning of the Motor Terminal boxes. ▪ **Quarterly** ➢ Checking of Power Cables termination during preventive Maintenance ▪ **Overhauling** ➢ Capacitance & tan delta of power cables are to be done and kept as reference.
06	Sluggish Operation of the Breaker	▪ **Overhauling** ➢ Inspection and Overhauling of the Breaker ➢ IR Value checking of the Breaker Pole to Pole and Incoming to Outgoing ➢ Testing of the Breaker (Contact Resistance, Breaker Timing)
07	Protective Devices Failure (Relay, Setting, Logic Deviation)	▪ **Daily** ➢ Visual inspection of the protective devices for any abnormal condition ➢ Visual Inspection of LEDs and resetting after due diligence ▪ **Overhauling** ➢ Feeders and Incomer settings are to be ensured as per approved and Recommended settings. Ensuring the implementation of guidelines issued by COS with respect to protection setting, Logic and configuration. ➢ Secondary injection testing of the protective devices is to be carried out including the measurement circuit. The output circuit shall be checked to ensure there is no impact on the other running system. ➢ Functional checks of each of the protective devices shall be ensured. ➢ Protection tripping of breakers are to be ensured. ➢ For Motor Feeders additionally the EPB tripping shall be checked ➢ For Transformer feeders Buchholz, PRV, OTI and WTI tripping is to be ensured. ➢ Activation of the Reverse blocking

		signals for the Incomer/Tie/Bus Coupler Breaker shall be ensured during feeder relay test. ➢ SOE and Alarms of the protection system and the Breakers shall be ensured as per LMI: OD-ELEC-003 ➢ Adherence to TC document (OGN/ELEC-040) issued on maintenance of numerical relays. ➢ All protection function, which are not required shall be disabled. ➢ Detail Point to point checking is to be ensured during first commissioning/R&M
08	(Others) Feedback Status	▪ **Overhauling** ➢ Functional Checks are to be done for each of the Breaker (ON, OFF, Protection Operated) ➢ Latch Relay is to be used for the Breaker Status to prevent loss of Contact during switchgear control supply disturbance.
09	Human Error	➢ While working on the Live panel/system (secondary circuit), SOP & JSA are to be prepared. ➢ The matter is to be discussed with senior people before executing the work
10	Moisture & Foreign Object Entry	▪ **Monthly** ➢ Inspection of the Breathers provided in the Bus duct (replacement of silica gel if required) ▪ **Overhauling** ➢ All cable Gland holes in the switchgears are to be sealed. ➢ Switchgear Rooms are to be sealed to prevent entry of moisture, Dust and Foreign objects. ➢ The Air washer system is to be ensured for proper operation.

11	220V DC Earth Fault	▪ **Daily** ➢ Monitoring of 220V DC Voltage in All DCDB (Positive to Earth & Negative to Earth). For ensuring that there is no DC Earth Fault. If any is to be identified and rectified at the earliest. ▪ **Quarterly** ➢ Ensuring DC Earth Fault signal to alarm/SOE in DCS. ➢ Ensure the DC voltages (P-E & N-E) are available in trends at DCS. ➢ DC fail shall lead to only stay put condition. Numerical relays after DC OFF shall lead to all output contact in original position so same to be judiciously used.

14. LIST OF MANDATORY SPARES FOR 33 KV/11KV/3.3KV SWITCHGEARS

14.1 33KV SWITCHGEARS

Sl No	Item Description	Minimum required Quantity
1	Breaker of each rating	1 no. breaker of eachtype and rating
2	Spring charging motor complete	2 Nos. of each type &rating
3	Shunt trip coil	2 Nos. of each type &rating
4	Closing coils	2 Nos. of each type &rating
5	Current transformer of each type & ratio	3 Nos. of each type
6	Potential transformer of each type & ratio	3 Nos. of each type
7	Numerical relays	1 Nos. of each type
8	Aux. Relays/ Lock out relays/ Timers	2 Nos. of each type
9	Moving contact assembly of each rating	1 Set
10	Stationary (fixed) contact of each rating.	1 Set
11	Bus seal off bushings	3 nos. of each type &size
12	Bus bar support insulators	3 Nos.
13	Limit switches	6 Nos.
14	Closing spring	2 Nos. of each type
15	Tripping spring	2 Nos. of each type
16	Control switches	2 Nos. of each type
17	Selector switches	2 Nos. of each type
18	Isolation switch for the control supply	2 Nos.
19	operating mechanism rod for each rating	1 set
20	Energy meter of each type and range	1 Nos. of each type

21	Circuit breaker auxiliary contact assembly	2 Nos. of each type &rating
22	Indicating lamps with holders	10 set
23	Fuse base and holder	10 Nos.
24	Fuse Link	10 Nos.
25	Isolating contact (fixed & moving) (one set means male & female contacts of one complete breaker)	1 set of each rating
26	Terminal blocks	2 Nos.
27	SF6 cylinder with SF6 gas filled + nozzle for filling thegas (if applicable)	
28	Multiple pin plug contact assy. with cables (male &female)	3 Nos.
29	Guide for moving contact set (one set means completereplacement for one breaker)	1 set
30	Interphase barrier	2 Nos. of each type
31	Pressure gauge (for SF6 breakers)	2 Nos.
32	Vacuum Bottle (For Vacuum breaker)	2 Sets

14.2 11KV/6.6KV/3.3KV SWITCHGEARS

Sl No	Item Description	Minimum required Quantity
1	Breaker of each rating	3 No of each type andrating
2	Spring charging motor complete	2 Nos. of each type
3	Shunt trip coil	5 Nos. of each type &rating
4	Closing coils	5 Nos. of each type &rating
5	Current transformer of each type & ratio	3 Nos. of each type
6	Potential transformer of each type & ratio	3 Nos. of each type
7	Numerical relays	2 Nos. of each type
8	Aux. Relays/ Lock out relays/ Timers/ Coupling relays (If applicable)	10 Nos. of each type

9	Moving contact assembly of each rating (One set means complete replacement for one breaker)	2 Set
10	Stationary (fixed) contact of each rating. (One set means complete replacement for one breaker)	2 Set
11	Bus seal off bushings	3 nos. of each type &size
12	Bus bar support insulators	12 Nos.
13	Limit switches of each type	10 Nos of each type
14	Closing spring	1 Set of each type
15	Tripping spring	1 Set of each type
16	Control switches	2 Nos. of each type
17	Selector switches	2 Nos. of each type
18	Isolation switch for the control supply (AC &DC)	2 Nos
19	operating mechanism rod for each rating	1 Set
20	Energy meter of each type & range	1 Nos. of each type
21	Circuit breaker auxiliary contact assembly	6 Nos. of each type &rating
22	Indicating lamps with holders (all colors to be covered)	20 Set
23	Control Fuse base and holder	10% of each type andrating
24	Control Fuse link	10% of each type andrating
25	Isolating contact (fixed & moving) (one set means male & female contacts of one complete breaker)	4 sets of each rating
26	Terminal blocks of each type	6 Nos of each type
27	SF6 Bottle	2 Sets of each rating
28	Multiple pin plug contact assy. with cables (male &female)	6 Nos
29	Interphase barrier	2 Nos. of each type
30	Vacuum Bottle	2 Sets of each rating

31	Surge arresters	3 Nos.
32	Vacuum Contactors with HRC fuses	2 Sets (One set forthree phases)
33	Maintenance Tools and accessories for maintenanceof HT Switchgear	2 Sets
34	Numerical Relays Networking System (Switchgear SCADA)	
34.1	Ethernet Switch of each type/model/ configuration (mounted in Switchgears/ data concentrator panels)	10% of total Quantity(Minimum 1No)
34.2	FO cable terminal equipment of each type such as LIUs along with associated patch cords and connectors	10% of total Quantity (Minimum 1No)
34.3	Power Supply modules of each type (mounted in Switchgears / data concentrator panels, as applicable)	10% of total Quantity(Minimum 1No)
34.4	Power Supply module for Server (DataConcentrator/HMI station)	4 Nos
34.5	Panel ventilation Fan	4 Nos
34.6	LAN Card for Server (Data concentrator/ HMI station)	4 Nos
34.7	Hard disc for Server (Data concentrator/ HMI station)	4 Nos
34.8	Power supply monitoring module / auxiliary relays as applicable)	4 Nos
34.9	Terminal connectors for FO/ Copper communicationcable	10 Nos of each type
34.10	Modules for Data Concentrator & system LAN	1 No of each type

15.0 LIST OF T&P ITEMS FOR HT SWITCHGEARS

Sl No	Item Description	Minimum required Quantity
1	Breaker timing test kit	1 Set
2	Contact resistance measurement kit	1 Set
3	CT analyzer test kit	1 Set
4	Relay secondary injection test kit	1 Set
5	Vacuum bottle testing kit	1 Set
6	IR test kit (From 250V to 5KV) digital	1 Set
7	IR test kit (From 500V to 15KV) digital	1 Set
8	Infrared thermography camera	1 Set
9	Primary injection kit suitable for testing of Current Transformers (Current range: 0-1000 amp.)	1 Set
10	Fully Automatic portable HT cable Fault Locator (cabletest kit)	1 Set
11	Analog Phase Sequence Indicator	1 Set
12	Crimping tools suitable for sizes: 0.5 sq.mm to 630 sq.mm)	1 Set
13	High sensitivity (able to detect high resistive faults), Precision grade Fully automatic DC earth fault locatorin live DC circuit.	1 Set
14	Precision grade Multi-meter (Digital)	1 no
15	Precision grade AC Clamp meter (up to 400A)	1 no
16	AC HiPOT Test Kit	1 Set
17	DC HiPOT Test Kit	1 Set

16.0 Recording Keeping:

Records will be kept in respective LMI file (HT Switchgear Section & Lab Group) of Electrical Maintenance.

17.0 Responsibility centers for implementation:

HOD (EMD) or his representative

18.0 Review:

HOD(EMD) will review & take the approval of competent authority (HOP). The document will be reviewed on 2 yearly basis or earlier ,if required.

19.0 STANDARD CHECKLISTS

19.1 STANDARD CHECKLIST FOR OVERHAULING OF HT SWITCHGEAR

Sl. No.	DESCRIPTION	Status (OK/NOT OK)
1	Issue of Permit to Work (PTW)	
2	Ensure proper isolation of the switchgear, Bus bar, Cables from both ends	
3	After isolation of the switchgear and before start of the work, a non-contact type of voltage detector is to be used for ensuring that the switchgear (where maintenance work is planned to be executed) is isolated from the live parts. If during isolation of any switchgear, the live part at some of the panel (like incomer/Tie) cannot be isolated, then proper locking arrangement of the live panel (from both front/rear side) is to be done with provision of danger tags, so that inadvertent opening of the panel is avoided	
4	Before opening of panel covers clean all the nearby areas	
5	Discharging of Bus bar and cables	
6	Checking of the IR value of Bus bar	
7	Cleaning of Panels, Bus bar Chamber, Cable chamber etc. (preferable by vacuum cleaner)	
8	Inspection of Bus bar joint & rectification & tightness by torque spanner	
9	Cleaning and physical inspection of all Bus Bar support insulators	
10	IR Value measurement of Bus bar after cleaning and blowing	
11	Milli volt drop measurement of all joints of the Bus bar	
12	Complete Bus bar Chamber box up	
13	Tightness checking of Power Cables	
14	Ensuring healthiness of space heater (if envisaged) in the cable compartment	

15	Care must be taken so that, the control/ CT/ PT secondary cables do not come in contact with the space heater (which maylead to tripping of the drive due to damage to the CT/ PT/ Control cables)	
16	Tightness checking of CT primary links and CT secondary cables at CT terminal. Check healthiness and tightness of equalizing potential lead if any.	
17	Ensure single point earthing of CT secondary side neutral connection	
18	Tightness checking of PT primary links, PT secondary cables atPT terminal and PT neutral earthing.	
19	Tightness checking of all CT Secondary, PT Secondary and Control cable terminals in Panel TB and in relay TB.	
20	It should be ensured that Ring type terminals are used at leastin the CT & PT circuit and disconnecting type of Links are used inCT terminals.	
21	Ensure healthiness of Switchgear earthing. Inspection and continuity of earthing is to be checked in all panel.	
22	Servicing and Testing of Breaker	
23	Checking of Electrical interlock function in service position (Busdead condition) for each breaker	
24	Alignment of Breaker Finger contacts and Bus bar contacts (for all breakers).Checking of contact overlap.	
25	Checking of proper functioning of Mechanical and Door interlock.	
26	Checking of functioning of Manual Emergency Trip lever in service position for each Breaker.	
27	Cleaning and physical inspection of Surge suppressor.	
28	Checking and cleaning of CVI (capacitive voltage insulator) if used for voltage indication or interlock purpose (monitoring of its voltage to be done during running for early detection of failure).	
29	Additional gland holes in the cable compartment should be checked and re- sealed. At certain places, if required, silicon sealant may be used to avoid entry of dust/moisture/rodent.	

30	Inspection of Panel/Door gaskets and replacement, if required. Check all the door locks are properly latched.	
31	If the cable trench is at the bottom of the switchgear, then it is to be ensured that there is no water accumulation in the trench and sealing of the trench is to be carried out to prevent water entry from outside.	
32	Checking of all exhaust vents of switchgear are cleaned and covered/canopy provided. There should not be any duct/opening above such vents. Proper working of Ventilation system is to be ensured.	

19.2 STANDARD CHECKLIST FOR OVERHAULING OF HT BREAKERS

Sl. No.	DESCRIPTION	Status (OK/NOT OK)
1	Cleaning and blowing of the Breaker	
2	Greasing of Gear box	
3	Cleaning & lubrication of ON/OFF coil plungers	
4	Tightness checking of connection inside breaker	
5	Condition of Multi pin socket	
6	Lubrication of all moving parts of Breaker, Fixed and moving contacts	
7	General cleaning and lubrication of draw out rail	
8	Resistance measurement of Closing Coil & replace if deviated	
9	Resistance measurement of Tripping Coil & replace if deviated	
10	Resistance measurement of spring charging Motor & replace if deviated	
11	Healthiness of Beaker auxiliary contacts and Limit switches	
12	Checking of Manual operation of Breaker, i.e. Spring charge, Close and Trip	
13	Electrical operation of Breaker	

19.3 STANDARD CHECKLIST FOR SWITCHGEAR CHARGING

Sl. No.	DESCRIPTION	Status (OK/NOTOK)
1	Ensure that all men, materials, and machines including local earthing isremoved from the switchgear, which is to be charged.	
2	Ensure that all the breakers in the switchgear (all feeders in the Bus)are in racked out & in OFF condition.	
3	CT secondary terminals are not in open condition	
4	All panel doors are in closed condition (control/Breaker and cable chamber in the Front side as well as in the back side)	
5	Measure the IR Value of the Bus just before the charging	
6	PT trolley is/are in raked in condition & secondary terminal are not open	
7	Take clearance from all concerned (switchgear/ protection group)	
8	Ensure availability of both the DC supply source for the switchgear	
9	Ensure all the protective Devices of the Breaker to be closed for chargingthe switchgear are in reset condition.	
10	Ensure availability of DC supply for the Breaker control and the protectiveDevices (Breaker to be closed for charging the switchgear)	
11	Cancel the permit for initiating the action for normalization of the incomer/Tie/Bus coupler breaker for charging of the switchgear	
12	The Breaker (which is required for charging the switchgear) should be closed from Remote by Operation.	
13	After charging of the switchgear, wait for 5 to 10 minutes and then measure the Bus Voltage at the PT secondary terminals	

20.0 PROTOCOL TO BE SIGNED DURING OVERHAULING

20.1 PROTOCOL FOR SAFETY & GENERAL

Sl. No.	DESCRIPTION	Status (OK/NOT OK)
1.	Only authorized personnel should be allowed to work on the MV (3.3/6.6/11/33 kV) switchgear	
2.	Prior to the start of maintenance work on any switchgear, Job safety Analysis (JSA)is to be carried out for identification of the hazards	
3.	The isolation/normalization procedure and the job safety analysis are to be discussed in advance with the operation personnel (responsible for doing the isolation and normalization)	
4.	Arc flash suit and PPE must be worn during isolation and normalization of the MV switchgear	
5.	After isolation of the switchgear and before start of the work, a non-contact type of voltage detector is to be used for ensuring that the switchgear (where maintenance work is planned to be executed) is isolated from the live parts. If during isolation of any switchgear, the live part at some of the panel (like incomer/Tie) cannot be isolated, then proper locking arrangement of the live panel (from both front/rear side) is to be done with provision of danger tags, so that inadvertent opening of the panel is avoided	
6.	Proper identification markings on front and back side of panels to be done. Marking to include name of equipment, panel number and cross marking across entire switchgear. Cross marking will help to identify the correct panel cover.	
7.	Labelling of Switchgear Panels (both Front & Rear side), preferably fluorescent paint/sticker.	
8.	Marking in the Rear side of the panel by Zigzag fluorescent tape to avoid fixing of wrong panels on the rear side	
9.	Ensure Incomer/Bus Coupler/ Tie breaker live side marking on panel	
10.	Availability of Air Washer system. The discharge duct of the air washer system should not be directed towards the panel	
11.	Availability of Insulating mat/Paint of appropriate rating at both front and rear side of the panel	
12.	Proper illumination of the complete switchgear room is to be ensured	
13.	All the entry points (bus duct entry) of the switchgear rooms are sealed	

14.	Housekeeping of the switchgear Room preferably mopping or vacuuming only	
15.	Availability of the Fresh air/Exhaust Fan, if envisaged	
16.	Availability of the Fire detection system & appearance of Alarm in the respectivesystem (preferably in the DCS system)	
17.	Availability of appropriate type and no. of Fire extinguishers	
18.	Ensure sealing of extra holes/gland holes at all panels of the switchgear which are not in use	
19.	Labelling of the LED/protective devices for identifying the reason of tripping byoperating personnel	

20.2 PROTOCOL FORMAT FOR SOE & ALARM FOR INDIVIDUAL BREAKER ANDCOMMON SWITCHGEAR SECTION

NTPC LIMITED

Project: **Simhadri Super Thermal power station**

PROTOCOL FOR SOE/ALARM FOR INDIVIDUAL BREAKER CONTROL OF HT SWITCHGEAR

Type of Feeder: Motor/Transformer/Incomer/Tie/Bus Coupler
Date:
Switchgear
Detail:
Feeder Detail:

Sl No	Description	LVS Alarm	DCS SOE	Remarks
1	Incomer CB Close Status		✓	
2	Incomer CB Open Status		✓	
3	Incomer/Tie CB Close Command from DDC		✓	
4	Incomer/Tie CB Open Command from DDC		✓	
5	Feeder CB Close Status		✓	
6	Feeder CB Open Status		✓	
7	Electrical Protection Operated			
8	Feeder MCC Disturbance			
9	CB is not in remote			
10	CB Trip Circuit Supervision			
11	Relay Unhealthy			
12	EPB Operated Alarm in Motor Feeder			
13	Overload Alarm in Motor Feeder			

Checked By:	Checked By:	Checked By:
Signature:	Signature:	Signature:
Name:	Name:	Name:
Dept: EMD	Dept: C&I	Dept: Operation

NTPC LIMITED

Project: **Simhadri Super Thermal power station**

PROTOCOL FOR SOE/ALARM FOR COMMON SWITCHGEAR SECTION

Date:
Switchgear Detail:

Sl No	Description	LVS Alarm	DCS SOE	Remarks
1	UT-ST Bus Changeover Trouble	✓	✓	
2	Bus Under Voltage Status	✓	✓	
3	Bus No-Volt Status	✓	✓	
4	Reverse blocking Signal Active	✓	✓	
5	DC Supply-1/2 Fail	✓		
6	PT Fuse Fail			

Checked By:
Signature:
Name:
Dept: EMD

Checked By:
Signature:
Name:
Dept: C&I

Checked By:
Signature:
Name:
Dept: Operation

21.0 CT, PT & CIRCUIT BREAKER TEST FORMATS

NTPC LIMITED

Project: **Simhadri Super Thermal power station**
Ref:
Reference Standard:

CT CHECKS

Unit No: , Voltage Rating: Feeder
Detail: Test Date:
Instrument Used & No: Instrument Calibration date:

OBSERVATIONS

1.TURNS RATIO TEST

R-phase						
SI No	Applied Primary Current(A)	CT Secondary Current (A)	Measured Ratio	Reference Ratio	% Error	Remarks
1						
2						
3						
4						
Y-phase						
SI No	Applied Primary Current(A)	CT Secondary Current(A)	Measured Ratio	Reference Ratio	% Error	Remarks
1						
2						
3						
4						
B-phase						
SI No	Applied Primary Current(A)	CT Secondary Current (A)	Measured Ratio	Reference Ratio	% Error	Remarks
1						
2						
3						
4						

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

NTPC LIMITED

Project: **Simhadri Super Thermal power station**
Ref:
Reference Standard:

CT CHECKS

Unit No: , Voltage Rating:
Feeder Detail:
Test Date:
Instrument Used & No: Instrument Calibration date:

OBSERVATIONS

SI No	Applied Voltage (V)	Current Measured (mA)			Remarks
		R-phase CT	Y-phase CT	B-phase CT	
1					
2					
3					
4					
5					
6					

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

NTPC LIMITED

Project: **Simhadri Super Thermal power station**
Ref:
Reference Standard:

CT CHECKS

Unit No: , Voltage
Rating: Feeder Detail:
Test Date:
Instrument Used & No:
Instrument Calibration date:

OBSERVATIONS

1. IR Value Test (Primary Winding to Earth) Temp – ^{0}C

Sl. No.	Phases	Primary winding to earth, MΩ	Primary winding to Secondary winding, MΩ	Remarks
1	R-Phase			
2	Y-Phase			
3	B-Phase			

2. IR Value Test (Secondary Winding Core to Core and Core to Earth)

SL No	Phases	IR of Secondary Winding wrt Core and Earth, MΩ				Remarks
	R-phase	Core-1	Core-2	Core-3	Earth	
1	Core-1					
2	Core-2					
3	Core-3					
	Y-phase					
4	Core-1					
5	Core-2					
6	Core-3					
	B-phase					
7	Core-1					
8	Core-2					
9	Core-3					

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

TITLE	Guidelines on HT switchgear maintenance practices
LMI	LMI/OGN/OPS/ELC/044, Rev:0

NTPC LIMITED

Project: **Simhadri Super Thermal power station**
Ref:
Reference Standard:

CT CHECKS

Unit No: , Voltage
Rating: Feeder Detail:
Test Date:
Instrument Used & No:
Instrument Calibration
date:

OBSERVATIONS

1.SECONDARY WINDING RESISTANCE MEASUREMENT

Temp- ^{0}C

Sl No	Core	Resistance in Ω			Remarks
		R-phase	Y-phase	B-phase	
1	Core-1				
2	Core-2				
3	Core-3				

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

NTPC LIMITED

Project: **Simhadri Super Thermal power station**
Reference Standard:

PT CHECKS

Unit No: , Voltage
Rating: Bus Detail:
Test Date:
Instrument Used & No:
Instrument Calibration date:

OBSERVATIONS

1.IR Value Test (Primary Winding to Earth)

Temp – ^{0}C

Sl. No.	Phases	Primary winding to earth, MΩ	Remarks
1	R-Phase		
2	Y-Phase		
3	B-Phase		

2.IR Value Test (Secondary Winding to Earth)

Sl. No.	Phases	Secondary winding to earth, MΩ	Remarks
1	R-Phase		
2	Y-Phase		
3	B-Phase		

3. **IR Value Test (Primary Winding to Secondary Winding)**

Sl. No.	Phases	Primary winding to Secondary winding, MΩ	Remarks
1	R-Phase		
2	Y-Phase		
3	B-Phase		

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

NTPC LIMITED

Project:Simhadri Super Thermal power station
Reference Standard:

PT CHECKS

Unit No: , Voltage
Rating:Bus Detail:
Test Date:
Instrument Used & No:
Instrument Calibration
date:

OBSERVATIONS

1.TURNS RATIO TEST

R-phase

SI No	Applied Primary Voltage(V)	PT Secondary Voltage(V)	Measured Ratio	Reference Ratio	% Error	Remarks
1						
2						
3						

Y-phase

SI No	Applied Primary Voltage(V)	PT Secondary Voltage(V)	Measured Ratio	Reference Ratio	% Error	Remarks
1						
2						
3						

B-phase

SI No	Applied Primary Voltage(V)	PT Secondary Voltage(V)	Measured Ratio	Reference Ratio	% Error	Remarks
1						
2						
3						

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

NTPC LIMITED

Project:Simhadri Super Thermal power station
Reference Standard:

PT CHECKS

Unit No: ,
Voltage Rating:
Bus Detail:
Test Date:
Instrument Used & No:
Instrument Calibration date:

OBSERVATIONS

1.MAGNETISING CURRENT TEST

Sl No	Applied Voltage (V)	Current Measured (mA)			Remarks
		R-phase VT	Y-phase VT	B-phase VT	
1					
2					
3					
4					
5					
6					

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

NTPC LIMITED

Project:Simhadri Super Thermal power station
Ref:
Reference Standard:

PT CHECKS

Unit No: , Voltage
Rating: Bus Detail:
Test Date:
Instrument Used & No:
Instrument Calibration date:

OBSERVATIONS

1.PT PRIMARY FUSE RESISTANCE

SI No	Phase	PT primary Fuse resistance in Ω	Remarks
1	R-phase		
2	Y-phase		
3	B-phase		

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

NTPC LIMITED

Project:Simhadri Super Thermal power station
Ref:
Reference Standard:

CB CHECKS

Feeder Detail:
Sl No & Make of CB:
Test Date:
Instrument Used & No:
Instrument Calibration date:

OBSERVATIONS

1.CB CONTACT RESISTANCE MEASUREMENT

Sl No	Applied DC Current (Amp)	Phase	CB Contact resistance in μΩ	Remarks
1	100 Amp	R-phase		
2		Y-phase		
3		B-phase		

Note: CB should be in closed condition.

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

NTPC LIMITED

Project:Simhadri Super Thermal power station
Ref:
Reference Standard:

CB CHECKS

Feeder Detail:
Sl No & Make of CB:
Test Date:
Instrument Used & No:
Instrument Calibration date:

OBSERVATIONS

1.CB TIMING TEST

Sl No	Phase	Operation	CB Operation Time (milli sec)	Remarks
1	R-phase	Close		
		Open		
		Close-Open		
2	Y-phase	Close		
		Open		
		Close-Open		
3	B-phase	Close		
		Open		
		Close-Open		

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

NTPC LIMITED

Project:Simhadri Super Thermal power station
Ref:
Reference Standard:

CB CHECKS

Feeder detail:
Sl No and Make of CB:
Test Date:
Instrument Used & No:
Instrument Calibration date:

OBSERVATIONS

1.IR Value across Phase to Earth and Phase to Phase while CB is in closed Condition

Sl. No.	Phases	IR Value in GΩ	Remarks
1	R-ph to Earth		
2	Y-ph to Earth		
3	B-ph to Earth		
4	R-ph to Y-ph		
5	Y-ph to B-ph		
6	B-ph to R-ph		

2.IR Value across fixed and moving contacts while CB is in open condition

Sl. No.	Phases	IR Value in GΩ	Remarks
1	IR across R-ph		
2	IR across Y-ph		
3	IR across B-ph		

REMARKS: The test results are within / out of the permissible limit.

Test Engineer:

22.0 RELAY TEST FORMATS

22.1 TRANSFORMER FEEDER WITHOUT DIFFERENTIAL PROTECRTION

NTPC LIMITED

Project:Simhadri Super Thermal power station

Feeder Type: **Transformer Feeder Without Differential Protection**
Feeder Details:
Switchgear and Voltage Rating:
Relay Model & Make:
Test Date:
Testing instrument Model & Make: Instrument Calibration date:

1.MEASUREMENT CHECKS:

Current Input Terminals: TB-

Phase	Injected current (A)	Measured Current in Relay	Remarks
RN			
YN			
BN			
RYB			

Voltage Input Terminals: TB-

Phase	Injected Voltage (V)	Measured Voltage in Relay	Remarks
RN			
YN			
BN			
RYB			

Note: Metering Circuit check will be done by injecting Current and Voltage in metering circuit and Current, Voltage & Power will be checked in Meters, transducers etc.

2.Instantaneous over current protection (50)

Current Input Terminals: TB-

Phase	Set value (A)	Operated value (A) and time(ms)	Remarks
RN			
YN			
BN			
RYB			

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

3.Back up over current protection (51)

Phase	Set value (A) and time(ms)	Operated value (A) and time(ms)	Remarks
RN			
YN			
BN			
RYB			

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

4.Sensitive Earth fault protection (50N)

CBCT current input terminals: TB-

(i) Sensitive earth fault during charging and running

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

(ii) Sensitive earth fault during running
Protection function is active during running(when inrush(I2f/If)<=15%)

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

5.Standby Earth fault Protection (51N)
current input terminals: TB-

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Lockout (86) relay operated to DDCMIS	

6.Restricted earth fault protection (64R)
current input terminals: TB-

Set value (A)	Operated value			LED	Remarks
	Operated Voltage (V)	Operated current (A)	Operated Time (ms)		

7.Functional Checks:

Slno	Functional checks	Lockout (86) operated	CB tripped	Signal to DDCMIS	LED	Remarks
1	Buchholz					
2	PRV					
3	WTI					

4	OTI					
5	Standby E/F (If protection is realized in LV Side and Upstream CB tripping is through Binary Input)					
6	MCC disturbance					
7	Remaining relay binary inputs, binary outputs and LED					

8.CT Side and relay side resistance measurement

Protections	CT inputs	TB details	Resistance in CT side (Ohm)	Resistance in relay side (Ohm)	Remarks
50,51 protection	RN				
	YN				
	BN				
Metering	RN				
	YN				
	BN				
50N protection	CBCT				
51N protection	SBEF CT				
REF protection	REF CT				

Tested by: Checked by:

Signature: Signature:

Name: Name:
Designation: Designation:

22.2 Transformer Feeder with Differential protection

NTPC Limited

Project:Simhadri Super Thermal power station

Feeder Type: **Transformer Feeder With Differential Protection**
Feeder Details:
Switchgear and Voltage Rating: Relay
Model & Make:
Test Date:
Testing instrument Model & Make:
Instrument Calibration date:

1.TESTING OF DIFFERENTIAL RELAY

HV Side Current Input Terminal: TB-LV Side
Current Input Terminal: TB-

I.MEASUREMENT CHECKS OF DIFFERENTIAL RELAY

Windings	Phase	Injected current(A)	Measured current in relay (A)	Remarks
Winding-1 (HV side CT)	RN			
	YN			
	BN			
	RYB			
Winding-1 (LV side CT)	RN			
	YN			
	BN			
	RYB			

II.STABILITY CHECKS OF DIFFERENTIAL RELAY

Winding-1	Rated current (I1) injected (A)	Winding-2	Rated current (I2) injected (A)	Diff Current (Id in PU) in relay	Restraint current (Ir in PU) in relay
IR		Ir			
IY		Iy			
IB		Ib			

Check all currents including differential & restraining current in relay measurement tab and ensure Id=0 and Ir=1 or 2 depend upon make and Model of relay (i.e. Ir=1 in siemens relay as Ir= (I1+I2)/2 whereas Ir=2 in SEL Relay, Ir= (I1+I2))

Note: As per transformer vector group, HV and LV side angle in relay testing kit is to be maintained during secondary current injection.

III. DIFFERENTIAL PICK UP TEST

a. Three Phase pick up test

Windings	Phase	Calculated Current (A)	Operated Current (A)	Trip Time (ms)	Remarks
Winding-1					
Winding-2					

b. Three Phase pick up test

Phase	Calculated Current (A)	Operated Current (A)	Trip Time (ms)	Remarks
RN				
YN				
BN				

Note: Single-Phase pick-up test will be performed in relay in star side of transformer only.

IV. DIFFERENTIAL PICK UP TEST

HV Side	Set current(A)	Operated Current (A)	Trip Time (ms)	Remarks
RN				
YN				
BN				
RYB				

V. SLOPE (S1) TEST

Inject rated current to Winding-1 and Winding-2 with balanced angle through 3-phase relay testing kit, then increase the Winding-1 current (I1) in desired step and check for differential protection operation.

Phase	Winding-1 current (I1)	Winding-2 current (I2)	Diff relay reading (Id)	Diff relay reading (Ir)
RYB				

Calculated Slope= (Id/Ir) *100. Calculated slope should be equal to Set Slope (S1).

Similarly, the Slope test should be done at higher current depending upon setting of Is2 in relay. (If setting of Is2=3, then slope (s1) test should be done at less than 3 times of rated current)

VI. SLOPE (S2) TEST

Suppose setting of Is2 (Ibias for start of Slope2) =3.

Inject 3 times rated current to Winding-1 and Winding-2 with balanced angle through 3-phase relay testing kit, then increase the Winding-1 current (I1) in desired step and check for differential protection operation.

Phase	Winding-1 current (I1)	Winding-2 current (I2)	Diff relay reading (Id1)	Diff relay reading (Ir1)
RYB				

Inject 4 times rated current to Winding-1 and Winding-2 with balanced angle through 3-phase relay testing kit, then increase the Winding-1 current (I1) in step and check for differential protection operation.

Phase	Winding-1 current (I1)	Winding-2 current (I2)	Diff relay reading (Id2)	Diff relay reading (Ir2)
RYB				

Calculated Slope = ((Id2-Id1)/(Ir2-Ir1)) *100. Calculated slope should be equal to Set Slope (S2).

VII. SECOND HARMONIC BLOCKING TEST

- ✓ Connect the Current Output-1 & the Current Output-2 of 3-phase relay testing kitto Winding-1 of relay.
- ✓ In 3-Phase testing kit, Select Current O/p-1 to HV Side rated current and frequency=50Hz.
- ✓ Similarly Select Current O/p-2 to LV Side rated current and frequency=100Hz
- ✓ Inject the Current output-1&2 to winding-1 of relay. Ensure angle of CurrentOutput-1 & the Current Output-2 are balanced and same.
- ✓ In the kit, decrease the Current O/p-2 in desired step till the differentialprotection operation.

Current output-1 at 50Hz	Injected current (I1)	Current output-2 at 100Hz	Injected current(I2)
IR		Ir	
IY		Iy	
IB		Ib	

2nd harmonic current (%) =(I2/I1) *100.

Measured 2nd harmonic current should be equal to set value of 2nd harmonic current.

VII. FIFTH HARMONIC BLOCKING TEST

- Connect the Current Output-1 & the Current Output-2 of 3-phase relay testing kitto Winding-1 of relay.
- In 3-Phase testing kit, Select Current O/p-1 to HV Side rated current and frequency=50Hz.
- Similarly Select Current O/p-2 to LV Side rated current and frequency=250Hz.
- Inject the Current output-1&2 to winding-1 of relay. Ensure angle of CurrentOutput-1 & the Current Output-2 are balanced and same.

In the kit, decrease the Current O/p-2 in desired step till the differentialprotection operation

Current output-1 at 50Hz	Injected current (I1)	Current output-2 at 250 Hz	Injected current(I2)
IR		Ir	
IY		Iy	
IB		Ib	

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

2.MEASUREMENT CHECKS:

Current Input Terminals: TB-

Phase	Injected current (A)	Measured Current in Relay	Remarks
RN			
YN			
BN			
RYB			

Voltage Input Terminals: TB-

Phase	Injected Voltage (V)	Measured Voltage in Relay	Remarks
RN			
YN			
BN			
RYB			

Note: Metering Circuit check will be done by injecting Current and Voltage in metering circuitand Current, Voltage & Power will be checked in Meters, transducers etc

3.Instantaneous over current protection (50)

Current Input Terminals: TB-

Phase	Set value (A)	Operated value (A) and time(ms)	Remarks
RN			
YN			
BN			
RYB			

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

4.Back up over current protection (51)

Phase	Set value (A) and time(ms)	Operated value (A) and time(ms)	Remarks
RN			
YN			
BN			
RYB			

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

5.Sensitive Earth fault protection (50N)

CBCT current input terminals: TB-

(iii) Sensitive earth fault during charging and running

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

(iv) Sensitive earth fault during running
Protection function is active during running(when inrush(I2f/If)<=15%)

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

6.Standby Earth fault Protection (51N)
current input terminals: TB-

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Lockout (86) relay operated to DDCMIS	

7.Restricted earth fault protection (64R)
current input terminals: TB-

Set value (A)	Operated value			LED	Remarks
	Operated Voltage (V)	Operated current (A)	Operated Time (ms)		

8.Functional Checks:

Slno	Functional checks	Lockout (86) operated	CB tripped	Signal to DDCMIS	LED	Remarks
1	Buchholz					
2	PRV					
3	WTI					

4	OTI					
5	Standby E/F (If protection is realized in LV Side and Upstream CB tripping is through Binary Input)					
6	MCC disturbance					
7	Remaining relay binary inputs, binary outputs and LED					

9.CT Side and relay side resistance measurement

Protections	CT inputs	TB details	Resistance in CT side (Ohm)	Resistance in relay side (Ohm)	Remarks
50,51 protection	RN				
	YN				
	BN				
Metering	RN				
	YN				
	BN				
50N protection	CBCT				
51N protection	SBEF CT				
REF protection	REF CT				

Tested by:

Signature:

Name:
Designation:

Checked by:

Signature:

Name:
Designation:

22.3 Motor Feeder Without Differential Protection

NTPC Limited

Project:Simhadri Super Thermal power station

Feeder Type: **Motor Feeder Without Differential Protection**
Feeder Details:
Switchgear and Voltage Rating: Relay
Model & Make:
Test Date:
Testing instrument Model & Make:
Instrument Calibration date:

1.<u>MEASUREMENT CHECKS:</u>

Current Input Terminals: TB-

Phase	Injected current (A)	Measured Current in Relay	Remarks
RN			
YN			
BN			
RYB			

Voltage Input Terminals: TB-

Phase	Injected Voltage (V)	Measured Voltage in Relay	Remarks
RN			
YN			
BN			
RYB			

2. over current protection (50)

Current Input Terminals: TB-

I. Over protection is active during both starting and running

Phase	Set value (A)	Operated value (A) and time(ms)	LED	Remarks
RN				
YN				
BN				
RYB				

II. Over protection is active during running

Over protection is active during both starting and running

Phase	Set value (A)	Operated value (A) and time(ms)	LED	Remarks
RN				
YN				
BN				
RYB				

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

3.Sensitive Earth fault protection (50N)

CBCT current input terminals: TB-

(v) Sensitive earth fault during charging and running

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

(vi) Sensitive earth fault during running

Protection function is active during running (when inrush(I2f/If)<=15%)

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	

Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

4.Load Jam protection

Current input terminal: TB-

Load jam protection is active during motor running

Phase	Set value (A)	Operated value (A) and time(ms)	LED	Remarks
RYB				

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Lockout (86) relay operated to DDCMIS	

5.Negative sequence current protection (46)

Current input terminal: TB-1

Phase	Set value (A) and Time(sec)	Operated value (A) and time(ms)	I2 current read in relay	LED	Remarks
RN					
YN					
BN					
RYB					

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Lockout (86) relay operated to DDCMIS	

6.Overload jam
Current input terminal: TB-1

Phase	Set value (A) and Time(sec)	Operated value (A) and time(ms)	LED	Remarks
RN				
YN				
BN				
RYB				

DC Logic Inputs

Description	Status
Thermal overload alarm to DDCMIS(TB-)	

7.Overload Trip
Current input terminal: TB-1

Phase	Set value (A) and Time(sec)	Operated value (A) and time(ms)	LED	Remarks
RN				
YN				
BN				
RYB				

DC Logic Inputs

Description	Status
Trip command to circuit breaker directly	

8.Under voltage Protection and PT fuse fail
CT Input Terminal: TB-
PT Input Terminal: TB-
Under voltage protection is operated with three phase undervoltage

Phase	Under voltage protection set value(V) and time delay (sec)	Operated value (V) and time(ms)	Status

Ensure the following during Under voltage check:

- ✓ Under Voltage Protection is blocked during PT Fuse Fail.
- ✓ During One Phase PT Fuse Failed, U< did not operate.
- ✓ During Two Phase PT Fuse Failed, U< did not operate.
- ✓ After two phase PT fuse Failed, 3rd phase PT fuse failed, U< did not trip.
- ✓ When simultaneously 3-phase undervoltage comes, then only after a delay of 2 Sec, Under Voltage protection operated and CB tripped.

DC Logic Inputs

Description	Status
Trip command to circuit breaker directly	
MCC disturbance (during PT fuse fail) (TB-) to DDCMIS	

9. Functional checks:

Slno	Functional Checks	Lockout(86) operated	CB tripped	Signal to DDCMIS	LED	Remarks
1	EPB Operated					
2	MCC Disturbance					
3	Remaining relay binary inputs, binary outputs and LED					

10. CT Side and relay side resistance measurement

Protections	CT inputs	TB details	Resistance in CT side (Ohm)	Resistance in relay side (Ohm)	Remarks
protection	RN				
	YN				
	BN				
Metering	RN				
	YN				
	BN				
50N protection	CBCT				

Tested by:

Signature:

Name:
Designation:

Checked by:

Signature:

Name:
Designation:

22.4 Motor Feeder With Differential Protection

NTPC Limited

Project:Simhadri Super Thermal power station

Feeder Type: **Motor Feeder With Differential Protection**
Feeder Details:
Switchgear and Voltage Rating: Relay
Model & Make:
Test Date:
Testing instrument Model & Make:
Instrument Calibration date:

1 Motor differential Protection (87)

There are two types of Differential Protection of Motor

(i) High Impedance Differential Protection
(ii) Low Impedance Differential Protection

A. HIGH IMPEDANCE DIFFERENTIAL PROTECTION

Current Input-1 Terminals: TB-
Current Input-2 Terminals: TB-

Open the Disconnecting link of the Current Input-1 & 2 in Switchgear Terminal Blocks. Inject the current through relay testing kit from Current input-1 or Current input-2.

Differential Protection during starting.

Phase	Set value(A)	Operated Voltage (V)	Operated Current (A)	LED
RN				
YN				
BN				

Differential Protection during runnig.

Phase	Set value(A)	Operated Voltage (V)	Operated Current (A)	LED
RN				
YN				
BN				

Stabilizing resistance = Ohm

<u>DC Logic Inputs</u>

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

B. LOW IMPEDANCE DIFFERENTIAL PROTECTION

Current Input-1 Terminals: TB-
Current Input-2 Terminals: TB-

I. Measurement checks of differential relay

Windings of relay	Phase	Injected current (A)	Measured current in relay (A)	Remarks
winding - 1	**RN**			
	YN			
	BN			
	RYB			
winding - 2	**RN**			
	YN			
	BN			
	RYB			

II. stability checks of differential relay

Winding-1	Rated current (I1) injected (A)	Winding-2	Rated current (I2) injected (A)	Diff Current (Id in PU) in relay	Restraint current (Ir in PU) in relay
IR		Ir			
IY		Iy			
IB		Ib			

Check all current including differential and restraining current in relay measurement tab and ensure Id =0 and IR =1or 2 depend upon make and model of relay(I.e Ir=1 in siemens relay as Ir= (I1+I2)/2 where as Ir=2 in SEL relay, Ir=(I1+I2)

III. DIFFERENTIAL PICK UP TEST

a. Three Phase pick up test

Windings	Phase	Calculated Current (A)	Operated Current (A)	Trip Time (ms)	Remarks
Winding-1	RN				
	YN				
	BN				
	RYB				
Winding-2	RN				
	YN				
	BN				
	RYB				

IV. SLOPE (S1) TEST

Inject rated current to Winding-1 and Winding-2 with balanced angle through 3-phase relay testing kit, then increase the Winding-1 current (I1) in desired step and check for differential protection operation.

Phase	Winding-1 current (I1)	Winding-2 current (I2)	Diff relay reading (Id)	Diff relay reading (Ir)
RYB				

Calculated Slope= (Id/Ir) *100. Calculated slope should be equal to Set Slope (S1).

Similarly, the Slope test should be done at higher current depending upon setting of Is2 in relay.(If setting of Is2=3, then slope (s1) test should be done at less than 3 times of rated current)

V. SLOPE (S2) TEST

Suppose setting of Is2 (Ibias for start of Slope2) =3.

Inject 3 times rated current to Winding-1 and Winding-2 with balanced angle through 3-phase relay testing kit, then increase the Winding-1 current (I1) in desired step and check for differential protection operation.

Phase	Winding-1 current (I1)	Winding-2 current (I2)	Diff relay reading (Id1)	Diff relay reading (Ir1)
RYB				

Inject 4 times rated current to Winding-1 and Winding-2 with balanced angle through 3-phase relay testing kit, then increase the Winding-1 current (I1) in step and check for differential protection operation.

Phase	Winding-1 current (I1)	Winding-2 current (I2)	Diff relay reading (Id2)	Diff relay reading (Ir2)
RYB				

Calculated Slope = ((Id2-Id1)/(Ir2-Ir1)) *100. Calculated slope should be equal to Set Slope (S2).

VI. SECOND HARMONIC BLOCKING TEST

a. Connect the Current Output-1 & the Current Output-2 of 3-phase relay testing kitto Winding-1 of relay.

- ✓ In 3-Phase testing kit, Select Current O/p-1 to HV Side rated current and frequency=50Hz.
- ✓ Similarly Select Current O/p-2 to LV Side rated current and frequency=100Hz
- ✓ Inject the Current output-1&2 to winding-1 of relay. Ensure angle of Current Output-1 & the Current Output-2 are balanced and same.
- ✓ In the kit, decrease the Current O/p-2 in desired step till the differential protection operation.

Current output-1 at 50Hz	Injected current (I1)	Current output-2 at 100Hz	Injected current(I2)
IR		Ir	
IY		Iy	
IB		Ib	

2^{nd} harmonic current (%) =(I2/I1) *100.

Measured 2^{nd} harmonic current should be equal to set value of 2^{nd} harmonic current.

VII. FIFTH HARMONIC BLOCKING TEST

a. Connect the Current Output-1 & the Current Output-2 of 3-phase relay testing kitto Winding-1 of relay.
b. In 3-Phase testing kit, Select Current O/p-1 to HV Side rated current and frequency=50Hz.
c. Similarly Select Current O/p-2 to LV Side rated current and frequency=250Hz.
d. Inject the Current output-1&2 to winding-1 of relay. Ensure angle of Current Output-1 & the Current Output-2 are balanced and same.

In the kit, decrease the Current O/p-2 in desired step till the differential protection operation

Current output-1 at 50Hz	Injected current (I1)	Current output-2 at 250Hz	Injected current(I2)
IR		Ir	
IY		Iy	
IB		Ib	

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

Note : plotting of differential protection characteristic should be done in relay testing software and differential protection testing should be done

2.MEASUREMENT CHECKS:

Current Input Terminals: TB-

Phase	Injected current (A)	Measured Current in Relay	Remarks
RN			
YN			
BN			
RYB			

Voltage Input Terminals: TB-

Phase	Injected Voltage (V)	Measured Voltage in Relay	Remarks
RN			
YN			
BN			
RYB			

3. over current protection (50)

Current Input Terminals: TB-

I. Over protection is active during both starting and running

Phase	Set value (A)	Operated value (A) and time(ms)	LED	Remarks
RN				
YN				
BN				
RYB				

II. **Over protection is active during running**

Over protection is active during both starting and running

Phase	Set value (A)	Operated value (A) and time(ms)	LED	Remarks
RN				
YN				
BN				
RYB				

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

4.Sensitive Earth fault protection (50N)

CBCT current input terminals: TB-

(vii) Sensitive earth fault during charging and running

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

(viii) Sensitive earth fault during running

Protection function is active during running(when inrush(I2f/If)<=15%)

Set value (A)	Operated value (A) and time(ms)	LED	Remarks

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated to DDCMIS	

5.Load Jam protection

Current input terminal: TB-

Load jam protection is active during motor running

Phase	Set value (A)	Operated value (A) and time(ms)	LED	Remarks
RYB				

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Lockout (86) relay operated to DDCMIS	

6.Negative sequence current protection (46)

Phase	Set value (A) and Time(sec)	Operated value (A) and time(ms)	I2 current read in relay	LED	Remarks

RN					
YN					
BN					
RYB					

Current input terminal: TB-1

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
Lockout (86) relay operated to DDCMIS	

7.Overload jam

Current input terminal: TB-1

Phase	Set value (A) and Time(sec)	Operated value (A) and time(ms)	LED	Remarks
RN				
YN				
BN				
RYB				

DC Logic Inputs

Description	Status
Thermal overload alarm to DDCMIS(TB-)	

8.Overload Trip

Current input terminal: TB-1

Phase	Set value (A) and Time(sec)	Operated value (A) and time(ms)	LED	Remarks
RN				
YN				
BN				
RYB				

DC Logic Inputs

Description	Status

Trip command to circuit breaker directly	

9.Under voltage Protection and PT fuse fail

CT Input Terminal: TB-

PT Input Terminal: TB-

Under voltage protection is operated with three phase undervoltage

Phase	Under voltage protection set value(V) and time delay (sec)	Operated value (V) and time(ms)	Status

Ensure the following during Under voltage check:

- ✓ Under Voltage Protection is blocked during PT Fuse Fail.
- ✓ During One Phase PT Fuse Failed, U< did not operate.
- ✓ During Two Phase PT Fuse Failed, U< did not operate.
- ✓ After two phase PT fuse Failed, 3rd phase PT fuse failed, U< did not trip.
- ✓ When simultaneously 3-phase undervoltage comes, then only after a delay of 2 Sec, Under Voltage protection operated and CB tripped.

DC Logic Inputs

Description	Status
Trip command to circuit breaker directly	
MCC disturbance (during PT fuse fail) (TB-) to DDCMIS	

10. Functional checks:

Slno	Functional Checks	Lockout(86) operated	CB tripped	Signal to DDCMIS	LED	Remarks
1	EPB Operated					
2	MCC Disturbance					
3	Remaining relay binary inputs, binary outputs and LED					

11. CT Side and relay side resistance measurement

Protections	CT inputs	TB details	Resistance in CT side (Ohm)	Resistance in relay side (Ohm)	Remarks
protection	RN				
	YN				
	BN				
Metering	RN				
	YN				
	BN				
50N protection	CBCT				

Tested by: Checked by:

Signature: Signature:

Name: Name:
Designation: Designation:

22.5 INCOMER/TIE/BUS COUPLER

NTPC Limited

Project:Simhadri Super Thermal power station

Feeder Type: **INCOMER/TIE/BUS COUPLER**
Feeder Details:
Switchgear and Voltage Rating: Relay
Model & Make:
Test Date:
Testing instrument Model & Make:
Instrument Calibration date:

1.<u>MEASUREMENT CHECKS:</u>

Current Input Terminals: TB-

Phase	Injected current (A)	Measured Current in Relay	Remarks
RN			
YN			
BN			
RYB			

Voltage Input Terminals: TB-

Phase	Injected Voltage (V)	Measured Voltage in Relay	Remarks
RN			
YN			
BN			
RYB			

2. over current protection (50)

Current Input Terminals: TB-

Phase	Set value (A)	Operated value (A) and time(ms)	LED	Remarks
RN				
YN				
BN				
RYB				

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
LED () indication for reverse blocking trip	
LED () indication for reverse input high	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated contact (TB-) to DDCMIS from incomer	

3.Residual Earth fault protection (50N)

Set value (A) T1(ms), T2 (ms)	**operated value (A)**	**T1 (ms)**	**T2 (ms)**	**LED**	**remarks**

Note:

(i)T2 time will be checked through reverse blocking contact simulation of one outgoing feeder of that bus for Incomer and Tie.

(ii) T2 time will be checked through reverse blocking contact simulation of one out going feeder of each bus for Bus Coupler.

(iii) For 11kV Unit Incomer, 51NUT pick up contact is to be shorted and testing of E/F protection is to be done for operation of E/F protection.

(iv) For 11kV Unit Tie/ Station Incomer/Station Tie, 51NST pick up contact is to be shorted and testing of E/F protection is to be done for operation of E/F protection.

DC Logic Inputs

Description	Status
Lockout (86) relay operated	
Trip command to circuit breaker directly if lock out relay is not available	
LED () indication for reverse blocking trip	
LED () indication for reverse input high	
Reverse blocking contact to incomer/tie/bus coupler	
Lockout (86) relay operated contact (TB-) to DDCMIS from incomer	

4.Functional checks:

Slno	Functional Checks	Lockout(86) operated	CB tripped	Signal to DDCMIS	LED	Remarks
1	MCC Disturbance					
2	Remaining relay binary inputs, binary outputs and LED					

5.CT Side and relay side resistance measurement

Protections	CT inputs	TB details	Resistance in CT side (Ohm)	Resistance in relay side (Ohm)	Remarks
50,51 protection	RN				
	YN				
	BN				
Metering	RN				
	YN				
	BN				

Tested by: Checked by:

Signature: Signature:

Name: Name:
Designation: Designation: